MIL-C-46927A (MU)
30 November 1971
SUPERSEDING
MIL-C-46927 (MU)
1 February 1963

MILITARY SPECIFICATION

CARTRIDGE, 90MM, PRACTICE, M371 WITH FUZE, PIBD, M530A1 AND MOX-2B PELLET LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of practice cartridge designated as Cartridge, 90MM, Practice, M371 for 90MM Rifle, M67.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-1000	-	Drawings, Engineering and Associated Lists
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY

8796716 - Box, Packing, Ammunition, For Cartridge, 90MM, M371 and M371A1 for Rifle, M67, in Fiber Container, PA56

8796717 - Container, Ammunition, Fiber, PA56 for Cartridge, 90MM, M371 and M371A1 for Rifle, M67.

8865243 - Cartridge, 90MM, Practice, M371 w/Fuze, PIBD, M530A1 and MOX-2B Pellet for 90MM Rifle, M67, Loading Assembly and Marking Diagram

PUBLICATIONS

ARMY

ET - 8865243 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on Drawing (dwg.) 8875243 and with all requirements specified in applicable specification.

3.3 Profile and alignment.-The cartridge shall freely enter a complete round profile and alignment gage.

3.4 Electrical circuit resistance.-The loaded and fuzed projectile, prior to assembly into the complete round, shall have an electrical circuit resistance of not less than 65,000 ohms, nor more than 300,000 ohms when tested as specified in 4.3.2.

3.5 Proving Ground.-The cartridge shall comply with the following requirements when tested as specified in 4.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

3.5.1 Muzzle velocity.-The corrected muzzle velocity at ambient temperature shall be not more than 730 feet per second (fps) nor less than 670 fps, and the standard deviation of the lot shall not exceed 10 fps.

3.5.2 Safety.-There shall be no premature burst or indication of metal parts failure in rifle bore or in flight before impact.

3.5.3 Functioning.-There shall be visible evidence of functioning upon impact with the target as indicated by presence of flash or smoke cloud.

3.6 Workmanship

3.6.1 All parts shall be free of chips, dirt, grease, rust and other foreign materials. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.6.1.1 There shall be no metal filing or chips present inside the nose cap.

3.6.2 The repainting and marking shall be performed by the assembling contractor in accordance with the requirements of applicable drawings. In addition, such painting and marking shall be restored when marred or obliterated during assembly.

3.7 Torque requirements.-Thread joints shall withstand subsequent reapplication of the minimum (min.) assembly torque specified on the applicable drawing, without moving.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot", as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn, and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Each lot shall contain:

a. Cartridge cases of one type of metal from one manufacturer, in accordance with the same drawing and drawing revision, same specification and specification revision.

b. Projectile parts from one manufacturer, in accordance with the same drawing and drawing revision, same specification and specification revision.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

c. Propellant from not more than one lot.

d. Loaded fuzes of one interfix lot number from one manufacturer.

e. Loaded primers of one interfix lot number from one manufacturer.

f. Inert Filler E of one interfix lot number from one manufacturer.

g. Pellet of one interfix lot number from one manufacturer.

h. Lucky assemblies from one manufacturer in accordance with the same drawing and drawing revision.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect, and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile loading assembly, prior to loading (See dwg. 8865242, covering a detail of dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Foreign matter in charge cavity	Visual	01001
202.	Interior protective coating with bare spot	Visual	01002

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.2.2.2 Pellet (See dwg. 8865241, covering a detail of dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Thickness	Gage	02001
102.	Diameter	Gage	02002
Minor:	None defined.		

4.2.2.3 Projectile loading assembly, prior to assembly of fuze shock pad (See dwg. 8865242, covering a detail of dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Depth of fuze cavity	Gage	03001
202.	Foreign matter in charge	Visual	03002

4.2.2.4 Projectile loading assembly, prior to assembly of fuze (See dwg. 8865242, covering a detail of dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pellet missing	Visual	04001
102.	Fuze shock pad missing	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	04003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.2.2.5 Loading assembly prior to assembly of adapter (See dwg. 8865242, covering a detail of dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Wire damaged	Visual	05001
102.	Terminal insecurely crimped to wire	Visual-Manual	05002
103.	Terminal insecurely fastened to fuze	Visual-Manual	05003
104.	Wave washer missing	Visual	05004
105.	Disc missing	Visual	05005
106.	Threads obviously malformed (See 6.4)	Visual	05006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	05007

4.2.2.6 Adapter, prior to assembling to loading assembly (See dwg. 8865242, covering a detail of dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Threads obviously malformed (See 6.4)	Visual	06001
Minor:	AQL 0.65 percent		
201.	Silicone compound missing from adapter threads	Visual	06002

4.2.2.7 Loading assembly prior to assembling fin (See dwg. 8865242, covering a detail of dwg. 8865243).

Critical:			
1.	Gap at adapter to body joint (See 6.5)	Gage	07001

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Major:	AQL 0.40 percent		
102.	Adapter fails to withstand subsequent reapplication of min. assembly torque (See 3.7 and 4.3.4).....	Test	07002

Minor: None defined.

4.2.2.8 Propelling charge and fin, prior to assembling to loading assembly (See dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Propelling charge bag damaged to the extent that propellant can escape.....	Visual	08001
Minor:	AQL 1.5 percent		
201.	Propelling charge bag damaged or stitches missing, but not to the extent that propellant can escape.....	Visual	08002
202.	Silicone compound missing from fin threads.....	Visual	08003
203.	Evidence of poor workmanship (See 3.6).....	Visual	08004

4.2.2.9 Projectile loading assembly, prior to assembly into cartridge case (See dwg. 8865242).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	100 percent inspection		
101.	360 degrees (°) gap between fin and adapter.....	Gage	09001
Major:	AQL 0.40 percent		
102.	Fin fails to withstand subsequent reapplication of min. assembly torque (See 3.7 and 4.3.4).....	Test	09002
103.	Adhesive missing from surface of projectile at described area.....	Visual	09003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.5 percent		
201.	Protective coating with bare spot	Visual	09004
202.	Marking incorrect or unidentifiable	Visual	09005
203.	Evidence of poor workmanship (See 3.6)	Visual	09006

4.2.2.10 Rupture disc (See dwg. 8848566, covering a detail of dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.50 percent		
201.	Thickness	Gage	10001
202.	Outside diameter	Gage	10002
203.	Inside diameter	Gage	10003
204.	Evidence of poor workmanship (See 3.6)	Gage	10004

4.2.2.11 Cartridge, prior to assembling nose cap (See dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Paper insulating cup missing from lucky	Visual	11001
Major:	100 percent inspection		
101.	Lucky assembly missing	Visual	11002
102.	Two or more paper insulating cups on lucky	Visual	11003
Minor:	AQL 1.0 percent		
201.	Silicone compound missing from spike threads	Visual	11004
202.	Evidence of poor workmanship (See 3.6)	Visual	11005

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.2.2.12 Cartridge assembly (See dwg. 8865243).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Primer protrudes beyond rear of cartridge case	Gage	12001
Major:	AQL 0.65 percent		
101.	Primer fails to withstand subsequent reapplication of min. assembly torque (See 3.7 and 4.3.4)	Test	12002
102.	Cartridge case crimp missing	Visual	12003
103.	Nose cap fails to withstand reapplication of min. assembly torque (See 3.7 and 4.3.4)	Test	12004
104.	Lacquer missing from primer flange and edge of rupture disc	Visual	12005
Minor:	AQL 2.5 percent		
201.	Protective coating with bare spot	Visual	12006
202.	Marking incorrect or unidentifiable	Visual	12007
203.	Silicone compound missing from primer threads (prior to assembly to fin)	Visual	12008
204.	Excessive adhesive on projectile after assembly	Visual	12009
205.	Evidence of poor workmanship (See 3.6)	Visual	12010

4.2.2.13 Unsealed fiber container (See dwg. 8796717).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Nose support missing	Visual	13001

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

Minor:	AQL 2.5 percent		
201.	Cartridge improperly assembled in container	Visual	13002
202.	Glue or asphalt on round	Visual	13003
203.	Round cannot be removed by hand	Manual	13004
204.	Container cap cannot be removed by hand	Manual	13005
205.	Evidence of poor workmanship (See 3.6)	Visual	13006

4.2.2.14 Sealed fiber container (See dwg. 8796717).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Container damaged or cut through all asphalt layers	Visual	14001
102.	Tape incomplete or badly wrinkled	Visual	14002
103.	Metal end loose or distorted	Visual-Manual	14003
Minor:	AQL 2.5 percent		
201.	Marking missing, incorrect or unidentifiable	Visual	14004
202.	Gap between cover and body of container exceeding 1/8 inch	Visual	14005
203.	Tear tab length inadequate	Manual	14006
204.	Cuts, scuffs or gouges in outer layer	Visual	14007
205.	Contents loose	Manual	14008
206.	Evidence of poor workmanship (See 3.6)	Visual	14009

4.2.2.15 Sealed packing box (See dwg. 8796716).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Box damaged exposing contents	Visual	15001
102.	Hardware or strapping missing, broken or loose	Visual-Manual	15002

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

103.	DOD symbol missing or incorrect	Visual	15003
104.	Board broken	Visual	15004
Minor:	AQL 1.5 percent		
201.	Metallic seal missing, unsealed or improperly positioned	Visual	15005
202.	Marking missing, incorrect, or unidentifiable	Visual	15006
203.	Hardware or strapping improperly engaged or mislocated	Visual-Manual	15007
204.	Evidence of poor workmanship (See 3.6)	Visual	15008

4.2.3.1 Bullet pull (See dwg. 8865423), Major Defect, Code No. 12011.-The sampling plan for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Example B3. Code Letter F and an AQL of 1.5 percent shall be used. The test shall be performed as specified in 4.3.1 with equipment in accordance with 4.2.4.

4.2.3.2 Electrical circuit resistance and safety (See 3.5), Critical Defect, Code No. 09007.-This test shall be performed 100 percent. Any assembly that fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.2, using equipment in accordance with 4.2.4.

4.2.3.3 Profile and alignment (See 3.3), Major Defect, Code No. 12012.-The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to meet the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.4 Metal filings or chips inside nose cap (See 3.6.1.1), Major Defect, Code No. 12013.-The sampling plan and procedure for this test shall be in accordance with MIL-STD-1235, using a CSP-2 plan having an AQL of 0.40 percent and inspection level III. The test shall be performed as specified in 4.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.2.3.5 Proving ground.-The following tests shall be performed as specified in 4.3.5.

4.2.3.5.1 Muzzle velocity (See 3.5.1), Major Defect, Code No. 12014.

4.2.3.5.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement, the sampling plan for this test shall be in accordance with Standard MIL-STD-414. Section B, Part II, Example B3, Code Letter I and an AQL of 1.0 percent shall be used. In addition, if the standard deviation of the sample rounds multiplied by 0.75 exceeds the applicable requirement, the lot shall be rejected.

4.2.3.5.1.2 After three consecutive lots have met the criteria of 4.2.3.5.1.1, the sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414. Section B, Part II, Example B3, Code Letter F, and an AQL of 1.00 percent shall be used. In addition, if the standard deviation of the sample rounds multiplied by 0.65 exceeds the applicable requirement, the lot shall be rejected.

4.2.3.5.2 Functioning (See 3.5.3), Major Defect, Code No. 12015.

4.2.3.5.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement, twenty-five sample cartridges shall be tested as specified in 4.3.5.2 for functioning. If five or more sample cartridges fail the specified requirement, the lot shall be rejected.

4.2.3.5.2.2 After three consecutive lots have met the criteria of 4.2.3.5.2.1, ten sample cartridges shall be tested as specified in 4.3.5.2 for functioning. If three or more sample cartridges fail the specified requirement, the lot shall be rejected.

4.2.3.5.3 Extreme temperature firing for information, Code No. 12016.

4.2.3.5.3.1 Velocity and target-impact dispersion.-Twenty cartridges shall be selected from each of the first three production lots and fired for muzzle velocity and target-impact dispersion information at -40 degrees F (°F) (10 rounds) and 125°F (10 rounds). The test procedure shall be in accordance with 4.3.5.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.2.3.5.4 Premature bursts and security of parts (See 3.5.2), Critical Defect, Code No. 12017.-If any of the sample cartridges used in the test for muzzle velocity, functioning or extreme temperature firing fail to comply with the requirement for premature burst and security of parts, the lot shall be rejected.

4.2.3.5.5 Target-impact dispersion for information, Code No. 12018.-The sample cartridges used in the test for muzzle velocity and extreme temperature shall be used for this test. Testing shall be in accordance with 4.3.5.1.

4.2.3.5.6 Recoil momentum for information, Code No. 12019.-Beginning with the first lot and continuing for the first three lots, the cartridges fired in the test for muzzle velocity shall be used for this test. Testing shall be in accordance with 4.3.5.3.

4.2.3.6 Check test for deterioration of primers.-If the total elapsed time between the original acceptance of any primer lot and the assembly of that primer lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in the applicable specification immediately before the primer lot is assembled into the cartridges (See 6.6).

4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-8865423 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.2.4.2 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Bullet pull.-The security of the crimp of the cartridge case to the projectile shall be tested by applying a force in an axial direction at the rate of 0.100 to 0.125 inches per minute, inclusive.

4.3.2 Electrical circuit resistance.-The projectile, prior to assembly of lucky, shall be tested by means of the equipment specified in 4.2.4. The short circuit current of the circuit shall not exceed ten (10) microamperes. The electrical circuit resistance shall be determined between the socket bushing in the nose cavity and a bare portion of the ogive. Observation shall be made for compliance with the applicable requirement.

4.3.3 Inspection of metal filings or chips inside nose cap.-The nose cap shall be unscrewed from the spike and the paper insulating cup shall also be removed for visual inspection of presence of metal filings or chips in the cap, on the spike, or in the insulating cup. Rounds containing filings of 1/16 inch in length or greater in spike or nose cap area shall be considered as defective. If metal filings or chips are present in a sample round, regardless of the sizes of the filings or chips, the sample round shall not be returned to the production lot until removal or cleaning of the filings is complete. Nose caps are not to be disassociated from rounds during the inspection.

4.3.4 Torque testing.-The minimum specified assembly torque shall be applied in a tightening direction while holding the parts immediately adjacent to the joints being tested so as to avoid possible loosening of any other joints. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.3.5 Proving ground.-These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedures. The sample cartridge shall be fired from a gun for which the round is standard and observation made for compliance with all applicable requirements (See 6.9).

4.3.5.1 Velocity and target-impact dispersion.-The ammunition fired in this test shall be conditioned at a temperature of 70 ± 1°F in a test chamber for that length of time necessary for the thermocoupled round to reach the prescribed temperature prior to firing. The cartridge shall be fired at a cloth target placed at a distance of 300 yards from the muzzle of the gun. Vertical and horizontal hit pattern results shall be expressed as probable errors in mils and shall be recorded on the firing record for informational purposes only (See 6.8). Muzzle velocity shall be determined and recorded for each round.

4.3.5.2 Functioning.-The ammunition fired in this test shall be conditioned at a temperature of 70° ± 1°F in a test chamber for that length of time necessary for the thermocoupled round to reach the prescribed temperature prior to firing. The rounds shall be fired at a 1/2-inch thick plywood target placed 300 feet from the muzzle of the gun. Observation shall be made for conformance with the applicable requirement.

4.3.5.3 Recoil momentum (For information only).-Recoil momentum in pound-seconds shall be measured using a standard ballistic pendulum during the velocity phase for both the test cartridge and for standard reference cartridges. The recoil momentum shall be recorded for information only.

4.3.5.4 Extreme temperature firing for information.

4.3.5.4.1 Velocity and target-impact dispersion.-For this test, ten rounds shall be temperature conditioned at -40° ± 1°F and ten rounds shall be conditioned at 125 ± 1°F. prior to firing. The cartridge shall be fired at a cloth target placed at a distance of 300 yards from the muzzle of the gun. Muzzle velocity and target-impact dispersion data shall be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

4.3.5.4.2 Functioning.-For this test ten rounds shall be temperature conditioned at -40° + 1°F and ten rounds shall be conditioned at 125° + 1°F prior to firing. The rounds shall be fired at a 1/2-inch thick plywood target placed 300 feet from the muzzle of the gun. Functioning shall be recorded for information only.

4.3.5.5 Safety.-All rounds fired in the proving ground tests shall be observed for premature bursts and other apparent safety hazards.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with dwg. 8796717 with marking and sealing in accordance with dwg. 8796521.

5.2 Packing

5.2.1 Level A.-The container shall be packed in accordance with dwg. 8796716 with marking and sealing in accordance with dwg. 8796522.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (See 6.3.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA-1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment codes as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment, as defined in 6.3.3 and 6.3.4, will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the time of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five-digit Code Number contained in Section 4 of this specification, and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-A. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Obviously malformed threads.-The visual inspections called out for obviously malformed threads are intended to eliminate previously undetected conditions which may result in significant weakening of threaded joints. Included are such defects as double-threading, missing threads or portions thereof, stripped threads, and significant damage to thread crests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

6.5 Gap at adapter to body joint.-This inspection shall be made with the use of a .002 inch thick feeler gage. The gage shall not enter the joint except as follows: The gage may enter to a depth not exceeding .125 inch along continuous arc not exceeding 90° of the circumference.

6.6 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.7 Visual examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.8 Submission of test data and ammunition data cards.-In addition to the normal distribution of records, when the cartridge is procured by the U.S. Army Munitions Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-QA-A-A
and SMUPA-AD-E-A2
Dover, New Jersey 07801

6.9 Proving Ground Test

6.9.1 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

6.9.2 Combining of proving ground tests.-When the contractor for the cartridge is the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with the proving ground tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

of the cartridge to save expense, upon agreement between the procuring activity and the contractor. In all cases where the cartridge specification does not cover all of the proving ground tests specified for the component, the additional tests specified in the component specifications shall be conducted.

6.9.3 Test Summary: See next page.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1315-A-002

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

TEST	SAMPLE SIZE INITIAL (NOTE 1)	SAMPLE SIZE SUBSEQUENT	REQUIREMENTS	SAMPLING PARAGRAPH	TEST PROCEDURES PARAGRAPH
Muzzle Velocity	25	10	a. Muzzle Velocity (3.5.1)	4.2.3.5.1	4.3.5.1
			b. Safety (3.5.2)	4.2.3.5.4	4.3.5.5
			c. Target-Impact Dispersion (Info)	4.2.3.5.5	4.3.5.1
Functioning	25	10	a. Functioning (3.5.3)	4.2.3.5.2	4.3.5.2
			b. Safety (3.5.2)	4.2.3.5.4	4.3.5.5
Extreme Temperature Firing	20	N/A	a. Muzzle Velocity (Info)	4.2.3.5.3.1	4.3.5.4.1
			b. Target-Impact Dispersion (Info)	4.2.3.5.3.1	4.3.5.4.1
	20	N/A	c. Functioning (Info)	4.2.3.5.3.2	4.3.5.4.2
			d. Safety (3.5.2)	4.2.3.5.4	4.3.5.5

NOTES: 1. Until three (3) consecutive lots are accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:18Z
Check the source to verify that this is the current version before use.